

Introdução ao pacote **tidyverse** com a base de dados Palmer Penguins

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Carregando pacotes

```
# install.packages("tidyverse") # Instala o tidyverse
# install.packages("palmerpenguins") # Instala o pacote com o dataset dos penguins
# install.packages("visdat") # Instala um pacote para ver, graficamente, valores faltantes
# install.packages("GGally") # Instala um pacote para fazer pairplots

library(tidyverse)
library(palmerpenguins)
library(visdat)
library(GGally)

# Força a biblioteca a ser carregada do pacote palmerpenguins
penguins <- palmerpenguins::penguins
```

O que é o tidyverse?

O tidyverse é um ecossistema de pacotes para:

- manipulação de dados;
- visualização de dados;
- organização de dados.

Alguns pacotes importantes dentro do ecossistema

Pacote	Função
ggplot2	Visualização
dplyr	Manipulação
tidyr	Organização
readr	Importação
stringr	Manipulação de texto

Consulte a documentação aqui para mais informações.

O conjunto de dados Palmer Penguins

Contém medidas morfológicas de pinguins observados em ilhas da Antártida. Consulte aqui a documentação da base.

Variáveis importantes

Variável	Descrição
species	Espécie
island	Ilha
bill_length_mm	Comprimento do bico (mm)
bill_depth_mm	Profundidade do bico (mm)
flipper_length_mm	Comprimento da nadadeira (mm)
body_mass_g	Massa corporal (g)
sex	Sexo
Year	Ano

Primeira inspeção dos dados

Entendendo os dados

```
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island  bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int> <fct>
## 1 Adelie  Torgersen      39.1          18.7          181          3750 male
## 2 Adelie  Torgersen      39.5          17.4          186          3800 fema~
## 3 Adelie  Torgersen      40.3          18           195          3250 fema~
## 4 Adelie  Torgersen      NA           NA           NA           NA <NA>
## 5 Adelie  Torgersen      36.7          19.3          193          3450 fema~
## 6 Adelie  Torgersen      39.3          20.6          190          3650 male
## # i 1 more variable: year <int>
```

Repare que a estrutura de dados de `penguins` não é um `dataframe` mas sim um `tibble`, uma estrutura própria do `tidyverse`.

```
# Outra forma de ver os dados, útil quando tem muitas colunas
glimpse(penguins)
```

```
## Rows: 344
## Columns: 8
## $ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adelie,~
## $ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgersen, ~
## $ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, 42.~
## $ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, 20.~
## $ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186, 1~
```

```
## $ body_mass_g      <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, 425~
## $ sex              <fct> male, female, female, NA, female, male, female, male, N~
## $ year             <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2~
```

Nomes das variáveis

```
names(penguins)
```

```
## [1] "species"          "island"           "bill_length_mm"   "bill_depth_mm"
## [5] "flipper_length_mm" "body_mass_g"      "sex"              "year"
```

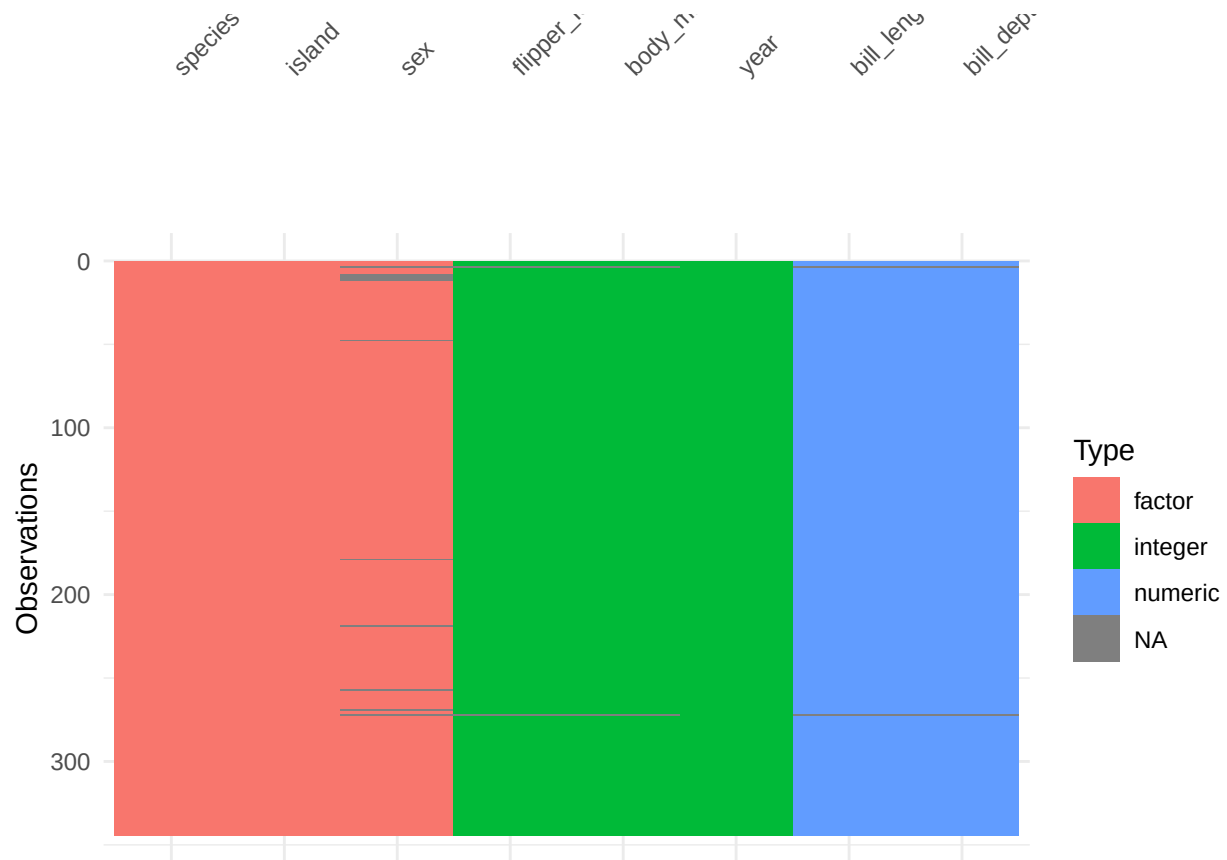
Resumo estatístico

```
summary(penguins)
```

```
##      species      island  bill_length_mm  bill_depth_mm  flipper_length_mm
## Adelie   :152  Biscoe   :168   Min.    :32.10   Min.    :13.10   Min.    :172.0
## Chinstrap: 68  Dream    :124   1st Qu.:39.23   1st Qu.:15.60   1st Qu.:190.0
## Gentoo   :124  Torgersen: 52   Median :44.45   Median :17.30   Median :197.0
##                                     Mean    :43.92   Mean    :17.15   Mean    :200.9
##                                     3rd Qu.:48.50   3rd Qu.:18.70   3rd Qu.:213.0
##                                     Max.    :59.60   Max.    :21.50   Max.    :231.0
##                                     NA's    :2      NA's    :2      NA's    :2
## body_mass_g      sex      year
## Min.    :2700   female:165   Min.    :2007
## 1st Qu.:3550   male  :168   1st Qu.:2007
## Median :4050   NA's   : 11   Median :2008
## Mean    :4202                                     Mean    :2008
## 3rd Qu.:4750                                     3rd Qu.:2009
## Max.    :6300                                     Max.    :2009
## NA's    :2
```

Visualizando a tipologia dos dados e dados faltantes

```
vis_dat(penguins)
```



Dados *tidy*

Três princípios fundamentais

1. Cada variável ocupa uma coluna e cada coluna é ocupada por uma variável (coluna = variável);
2. Cada observação ocupa uma linha e cada linha é ocupada por uma observação (linha = observação);
3. Cada valor ocupa uma célula e cada célula é ocupada por um único valor (célula = valor)

A base de dados `penguins` (bem como todas que trabalhamos no curso) já estão *tidy*.

Filtragens usando `dplyr`

`filter()`

Seleciona linhas com base em alguma condição.

```
penguins %>%
  filter(species == "Gentoo")
```

```
## # A tibble: 124 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
```

```
##      <fct>   <fct>           <dbl>           <dbl>           <int>           <int> <fct>
##  1 Gentoo   Biscoe           46.1           13.2             211           4500 female
##  2 Gentoo   Biscoe           50            16.3             230           5700 male
##  3 Gentoo   Biscoe           48.7           14.1             210           4450 female
##  4 Gentoo   Biscoe           50            15.2             218           5700 male
##  5 Gentoo   Biscoe           47.6           14.5             215           5400 male
##  6 Gentoo   Biscoe           46.5           13.5             210           4550 female
##  7 Gentoo   Biscoe           45.4           14.6             211           4800 female
##  8 Gentoo   Biscoe           46.7           15.3             219           5200 male
##  9 Gentoo   Biscoe           43.3           13.4             209           4400 female
## 10 Gentoo   Biscoe           46.8           15.4             215           5150 male
## # i 114 more rows
## # i 1 more variable: year <int>
```

Aceita múltiplas condições

```
penguins %>%
  filter(
    species == "Gentoo",
    body_mass_g > 5000
  )
```

```
## # A tibble: 61 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
##   <fct>   <fct>           <dbl>           <dbl>           <int>           <int> <fct>
##  1 Gentoo   Biscoe           50            16.3             230           5700 male
##  2 Gentoo   Biscoe           50            15.2             218           5700 male
##  3 Gentoo   Biscoe           47.6           14.5             215           5400 male
##  4 Gentoo   Biscoe           46.7           15.3             219           5200 male
##  5 Gentoo   Biscoe           46.8           15.4             215           5150 male
##  6 Gentoo   Biscoe           49            16.1             216           5550 male
##  7 Gentoo   Biscoe           48.4           14.6             213           5850 male
##  8 Gentoo   Biscoe           49.3           15.7             217           5850 male
##  9 Gentoo   Biscoe           49.2           15.2             221           6300 male
## 10 Gentoo   Biscoe           48.7           15.1             222           5350 male
## # i 51 more rows
## # i 1 more variable: year <int>
```

select()

Extraí colunas da base de dados.

```
penguins %>%
  select(species, island, body_mass_g)
```

```
## # A tibble: 344 x 3
##   species island body_mass_g
##   <fct>   <fct>           <int>
##  1 Adelie  Torgersen           3750
##  2 Adelie  Torgersen           3800
##  3 Adelie  Torgersen           3250
##  4 Adelie  Torgersen           NA
```

```
## 5 Adelie Torgersen 3450
## 6 Adelie Torgersen 3650
## 7 Adelie Torgersen 3625
## 8 Adelie Torgersen 4675
## 9 Adelie Torgersen 3475
## 10 Adelie Torgersen 4250
## # i 334 more rows
```

É possível selecionar um intervalo de colunas.

```
penguins %>%
  select(species:flipper_length_mm)
```

```
## # A tibble: 344 x 5
##   species island bill_length_mm bill_depth_mm flipper_length_mm
##   <fct>   <fct>         <dbl>         <dbl>         <int>
## 1 Adelie Torgersen      39.1          18.7          181
## 2 Adelie Torgersen      39.5          17.4          186
## 3 Adelie Torgersen      40.3           18          195
## 4 Adelie Torgersen      NA           NA           NA
## 5 Adelie Torgersen      36.7          19.3          193
## 6 Adelie Torgersen      39.3          20.6          190
## 7 Adelie Torgersen      38.9          17.8          181
## 8 Adelie Torgersen      39.2          19.6          195
## 9 Adelie Torgersen      34.1          18.1          193
## 10 Adelie Torgersen      42           20.2          190
## # i 334 more rows
```

Também é possível excluir colunas.

```
penguins %>%
  select(-island)
```

```
## # A tibble: 344 x 7
##   species bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex   year
##   <fct>         <dbl>         <dbl>         <int>         <int> <fct> <int>
## 1 Adelie      39.1          18.7          181          3750 male   2007
## 2 Adelie      39.5          17.4          186          3800 female 2007
## 3 Adelie      40.3           18          195          3250 female 2007
## 4 Adelie      NA           NA           NA           NA <NA>   2007
## 5 Adelie      36.7          19.3          193          3450 female 2007
## 6 Adelie      39.3          20.6          190          3650 male   2007
## 7 Adelie      38.9          17.8          181          3625 female 2007
## 8 Adelie      39.2          19.6          195          4675 male   2007
## 9 Adelie      34.1          18.1          193          3475 <NA>   2007
## 10 Adelie      42           20.2          190          4250 <NA>   2007
## # i 334 more rows
```

mutate()

Cria novas variáveis.

Exemplo: Peso em kg

```
penguins %>%  
  mutate(body_mass_kg = body_mass_g / 1000)
```

```
## # A tibble: 344 x 9  
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex  
##   <fct>   <fct>         <dbl>         <dbl>           <int>    <int> <fct>  
## 1 Adelie  Torger~         39.1          18.7             181      3750 male  
## 2 Adelie  Torger~         39.5          17.4             186      3800 fema~  
## 3 Adelie  Torger~         40.3           18             195      3250 fema~  
## 4 Adelie  Torger~          NA           NA              NA        NA <NA>  
## 5 Adelie  Torger~         36.7          19.3             193      3450 fema~  
## 6 Adelie  Torger~         39.3          20.6             190      3650 male  
## 7 Adelie  Torger~         38.9          17.8             181      3625 fema~  
## 8 Adelie  Torger~         39.2          19.6             195      4675 male  
## 9 Adelie  Torger~         34.1          18.1             193      3475 <NA>  
## 10 Adelie Torger~         42           20.2             190      4250 <NA>  
## # i 334 more rows  
## # i 2 more variables: year <int>, body_mass_kg <dbl>
```

Exemplo: Razão entre medidas do bico

```
penguins %>%  
  mutate(  
    bill_ratio = bill_length_mm / bill_depth_mm  
  )
```

```
## # A tibble: 344 x 9  
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex  
##   <fct>   <fct>         <dbl>         <dbl>           <int>    <int> <fct>  
## 1 Adelie  Torger~         39.1          18.7             181      3750 male  
## 2 Adelie  Torger~         39.5          17.4             186      3800 fema~  
## 3 Adelie  Torger~         40.3           18             195      3250 fema~  
## 4 Adelie  Torger~          NA           NA              NA        NA <NA>  
## 5 Adelie  Torger~         36.7          19.3             193      3450 fema~  
## 6 Adelie  Torger~         39.3          20.6             190      3650 male  
## 7 Adelie  Torger~         38.9          17.8             181      3625 fema~  
## 8 Adelie  Torger~         39.2          19.6             195      4675 male  
## 9 Adelie  Torger~         34.1          18.1             193      3475 <NA>  
## 10 Adelie Torger~         42           20.2             190      4250 <NA>  
## # i 334 more rows  
## # i 2 more variables: year <int>, bill_ratio <dbl>
```

arrange()

Ordena as observações (em ordem crescente) por alguma coluna.

```
penguins %>%
  arrange(body_mass_g)
```

```
## # A tibble: 344 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
##   <fct>   <fct>         <dbl>         <dbl>           <int>     <int> <fct>
## 1 Chinstr~ Dream          46.9           16.6             192       2700 fema~
## 2 Adelie   Biscoe          36.5           16.6             181       2850 fema~
## 3 Adelie   Biscoe          36.4           17.1             184       2850 fema~
## 4 Adelie   Biscoe          34.5           18.1             187       2900 fema~
## 5 Adelie   Dream          33.1           16.1             178       2900 fema~
## 6 Adelie   Torge~          38.6            17             188       2900 fema~
## 7 Chinstr~ Dream          43.2           16.6             187       2900 fema~
## 8 Adelie   Biscoe          37.9           18.6             193       2925 fema~
## 9 Adelie   Dream          37.5           18.9             179       2975 <NA>
## 10 Adelie  Dream          37            16.9             185       3000 fema~
## # i 334 more rows
## # i 1 more variable: year <int>
```

Em ordem decrescente:

```
penguins %>%
  arrange(desc(body_mass_g))
```

```
## # A tibble: 344 x 8
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex
##   <fct>   <fct>         <dbl>         <dbl>           <int>     <int> <fct>
## 1 Gentoo   Biscoe          49.2           15.2             221       6300 male
## 2 Gentoo   Biscoe          59.6            17             230       6050 male
## 3 Gentoo   Biscoe          51.1           16.3             220       6000 male
## 4 Gentoo   Biscoe          48.8           16.2             222       6000 male
## 5 Gentoo   Biscoe          45.2           16.4             223       5950 male
## 6 Gentoo   Biscoe          49.8           15.9             229       5950 male
## 7 Gentoo   Biscoe          48.4           14.6             213       5850 male
## 8 Gentoo   Biscoe          49.3           15.7             217       5850 male
## 9 Gentoo   Biscoe          55.1            16             230       5850 male
## 10 Gentoo  Biscoe          49.5           16.2             229       5800 male
## # i 334 more rows
## # i 1 more variable: year <int>
```

Os procedimentos acima não alteram a base original, ou seja, novas colunas não são acrescentadas na base nem sua ordenação natural é alterada. Para incorporar essas modificações, crie uma nova variável ou sobrescreva a variável original:

```
penguins2 <- penguins %>%
  mutate(body_mass_kg = body_mass_g / 1000)

names(penguins2)
```

```
## [1] "species"          "island"           "bill_length_mm"   "bill_depth_mm"
## [5] "flipper_length_mm" "body_mass_g"      "sex"              "year"
## [9] "body_mass_kg"
```


summarize()

Produz resumos estatísticos.

```
penguins %>%
  summarize(
    Média = mean(body_mass_g, na.rm = TRUE),
    DP = sd(body_mass_g, na.rm = TRUE),
    Min = min(body_mass_g, na.rm = TRUE),
    Max = max(body_mass_g, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 4
##   Média    DP    Min    Max
##   <dbl> <dbl> <int> <int>
## 1 4202.  802.  2700  6300
```

A grafia britânica, `summarise`, também funciona.

group_by()

1. Divide os dados em grupos;
2. Aplica uma operação;
3. Recombina os resultados.

Exemplo: Média por espécie

```
penguins %>%
  group_by(species) %>%
  summarise(
    media = mean(body_mass_g, na.rm = TRUE)
  )
```

```
## # A tibble: 3 x 2
##   species    media
##   <fct>    <dbl>
## 1 Adelie  3701.
## 2 Chinstrap 3733.
## 3 Gentoo  5076.
```

Exemplo: Média por espécie e ilha

```
penguins %>%
  group_by(species, island) %>%
  summarise(
    media = mean(body_mass_g, na.rm = TRUE)
  )
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by species and island.
## i Output is grouped by species.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(species, island))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.

## # A tibble: 5 x 3
## # Groups:   species [3]
##   species island   media
##   <fct>    <fct>   <dbl>
## 1 Adelie   Biscoe    3710.
## 2 Adelie   Dream     3688.
## 3 Adelie   Torgersen 3706.
## 4 Chinstrap Dream    3733.
## 5 Gentoo   Biscoe    5076.
```

Exemplo: Média considerando apenas uma espécie, agrupando por ilha

```
penguins %>%
  filter(species == "Adelie") %>%
  group_by(island) %>%
  summarise(media = mean(body_mass_g, na.rm = TRUE))
```

```
## # A tibble: 3 x 2
##   island   media
##   <fct>   <dbl>
## 1 Biscoe    3710.
## 2 Dream    3688.
## 3 Torgersen 3706.
```

Gráficos usando o ggplot2

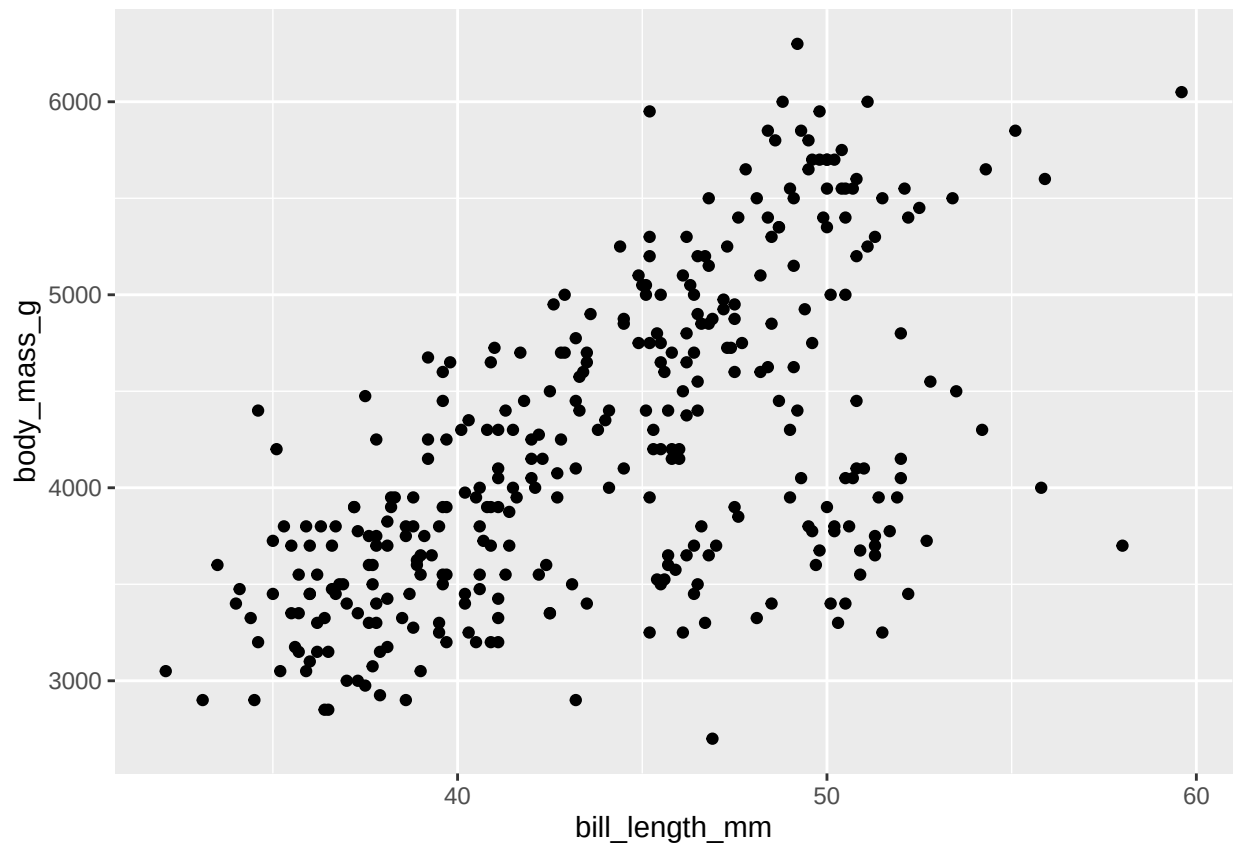
Da documentação do pacote: *Na maioria dos casos você começa com `ggplot()`, fornece um conjunto de dados e um mapeamento estético (com `aes()`). Em seguida, você adiciona camadas (como `geom_point()` ou `geom_histogram()`), escalas (como `scale_color_manual()`), especificações de facetas (como `facet_wrap()`) e sistemas de coordenadas (como `coord_flip()`).*

Vejamos diversos exemplos.

Gráfico de dispersão

```
ggplot(
  penguins,
  aes(
    x = bill_length_mm,
    y = body_mass_g
  )
) +
  geom_point()
```

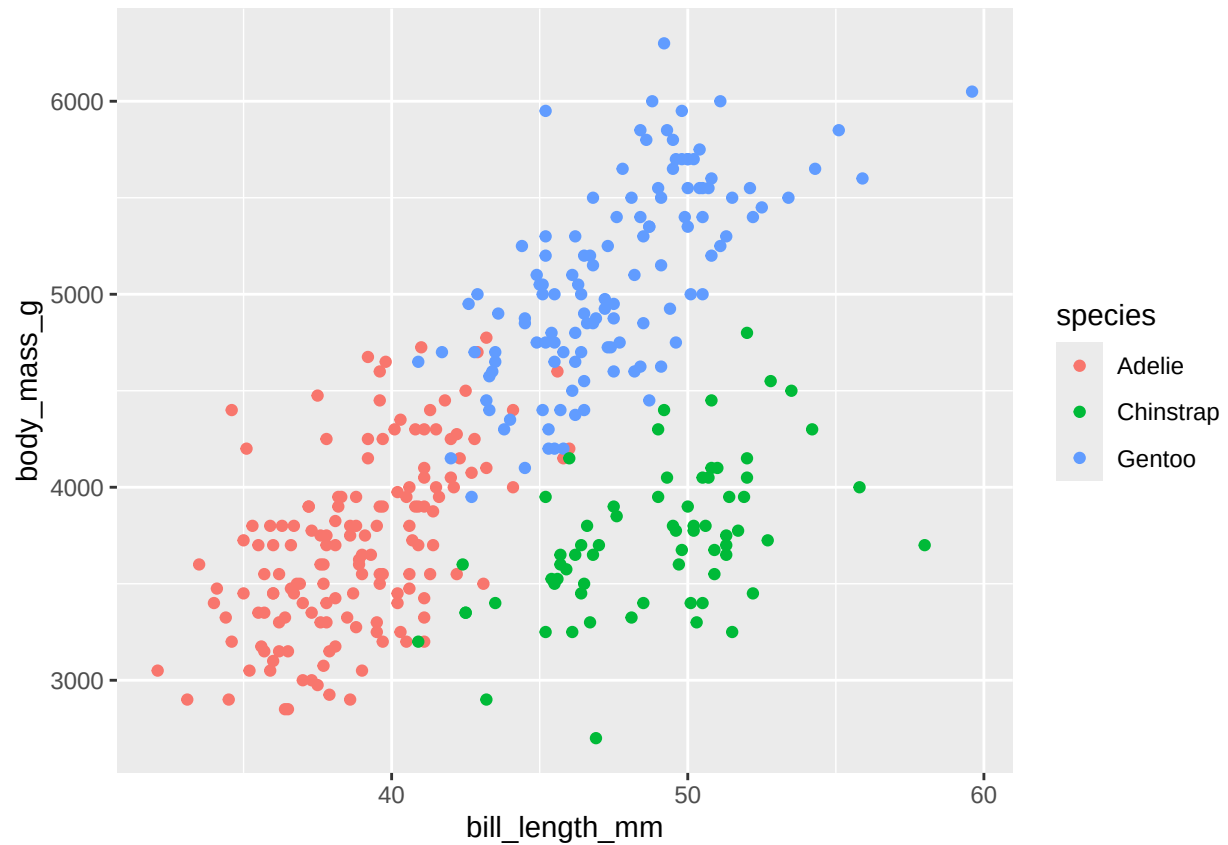
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Agora colorindo por espécie:

```
ggplot(  
  penguins,  
  aes(  
    x = bill_length_mm,  
    y = body_mass_g,  
    color = species  
  )  
) +  
  geom_point()
```

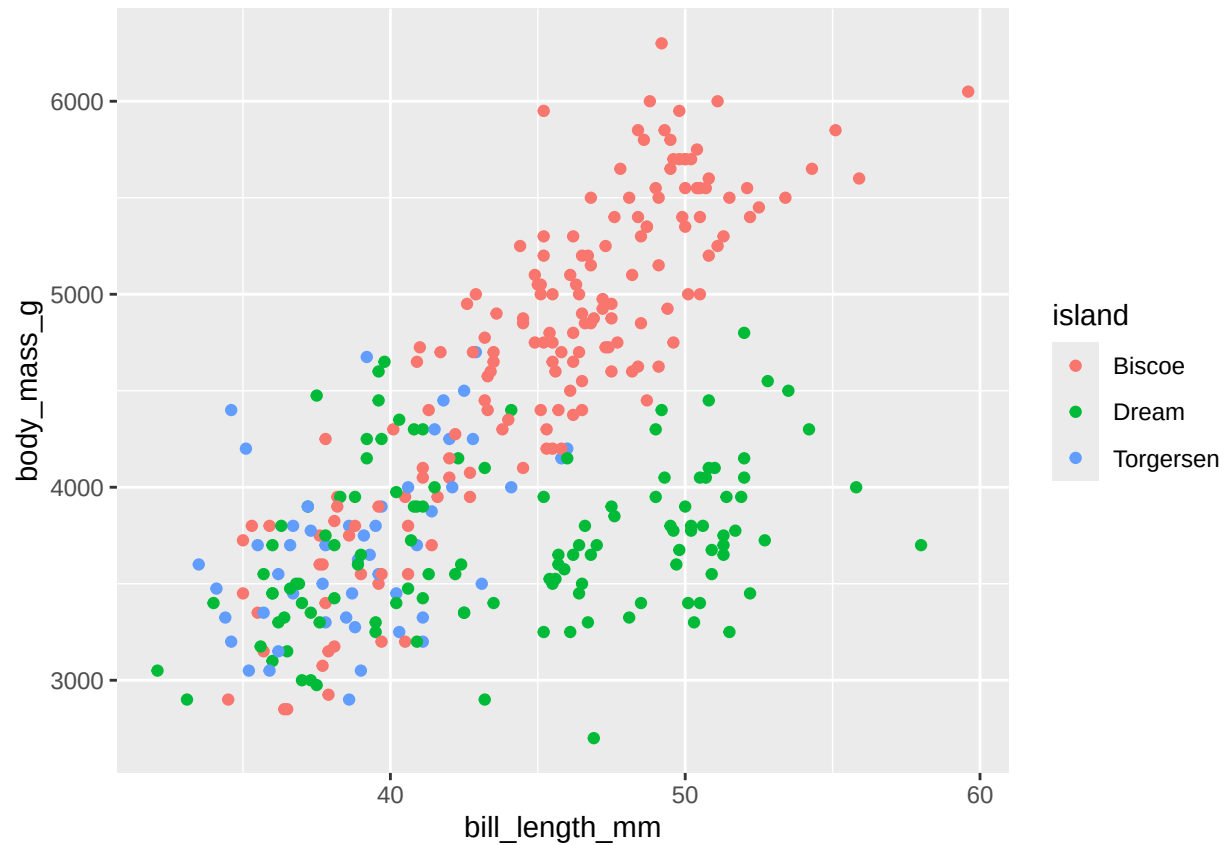
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Ou por ilha:

```
ggplot(  
  penguins,  
  aes(  
    x = bill_length_mm,  
    y = body_mass_g,  
    color = island  
  )  
) +  
  geom_point()
```

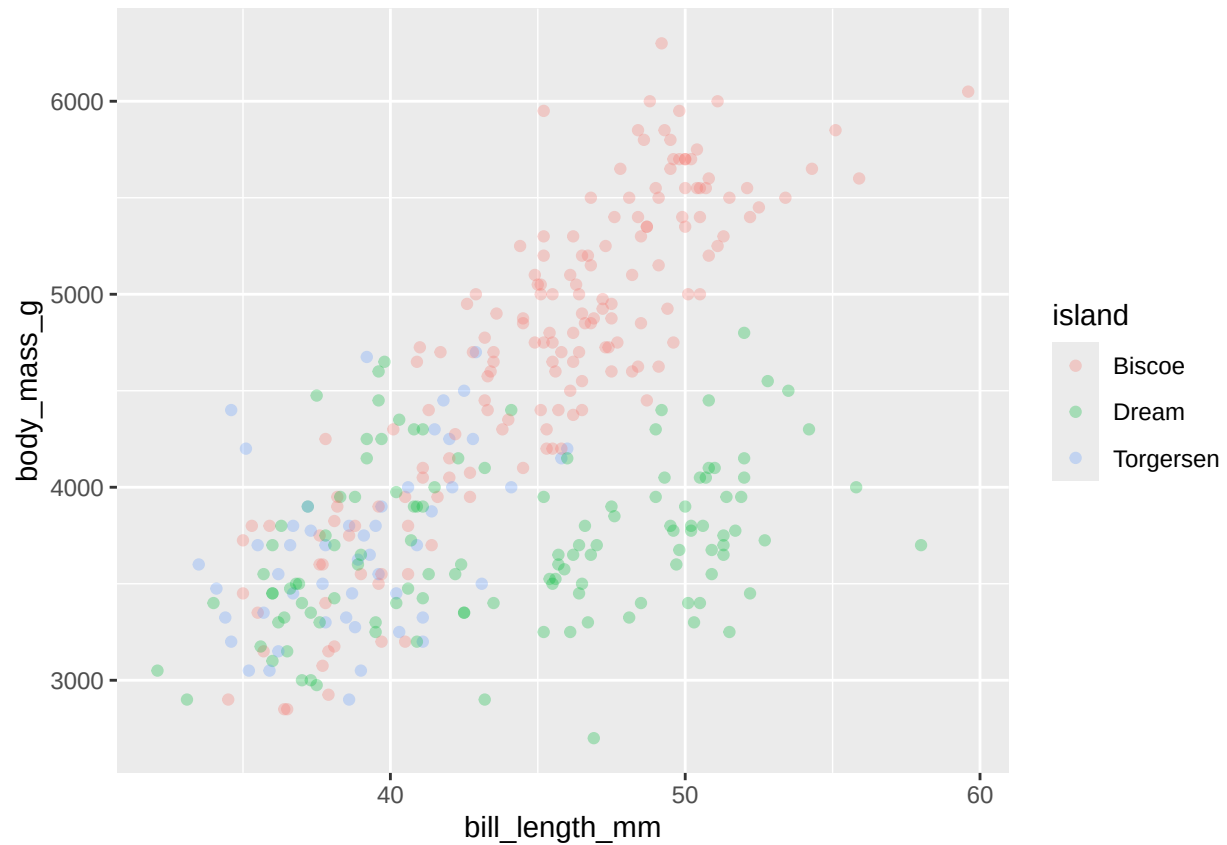
```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



Alterando a transparência:

```
ggplot(  
  penguins,  
  aes(  
    x = bill_length_mm,  
    y = body_mass_g,  
    color = island  
  )  
) +  
  geom_point(alpha = 0.3)
```

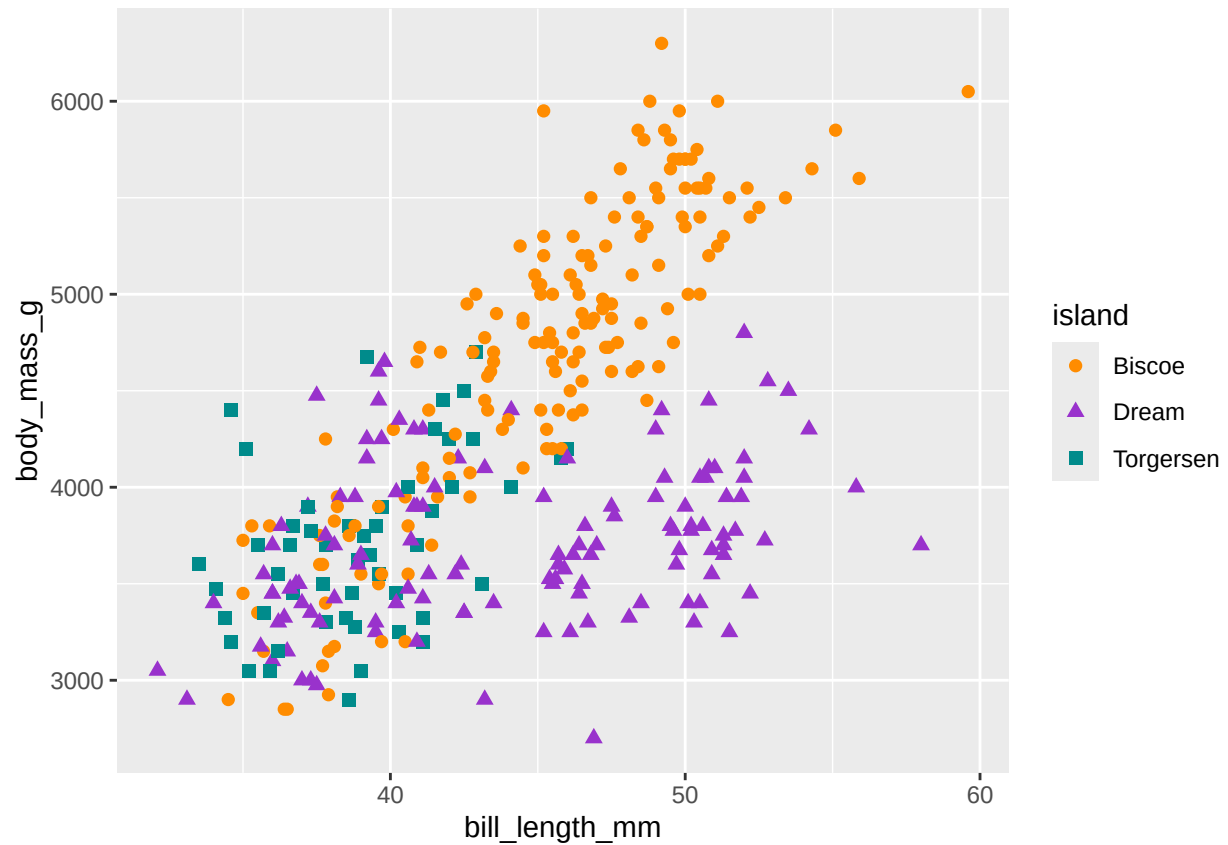
```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



Alterando mais coisas:

```
ggplot(  
  data = penguins,  
  aes(  
    x = bill_length_mm,  
    y = body_mass_g,  
    color = island,  
    shape = island  
  )  
) +  
  geom_point(size = 2) +  
  scale_color_manual(values = c("darkorange", "darkorchid", "cyan4"))
```

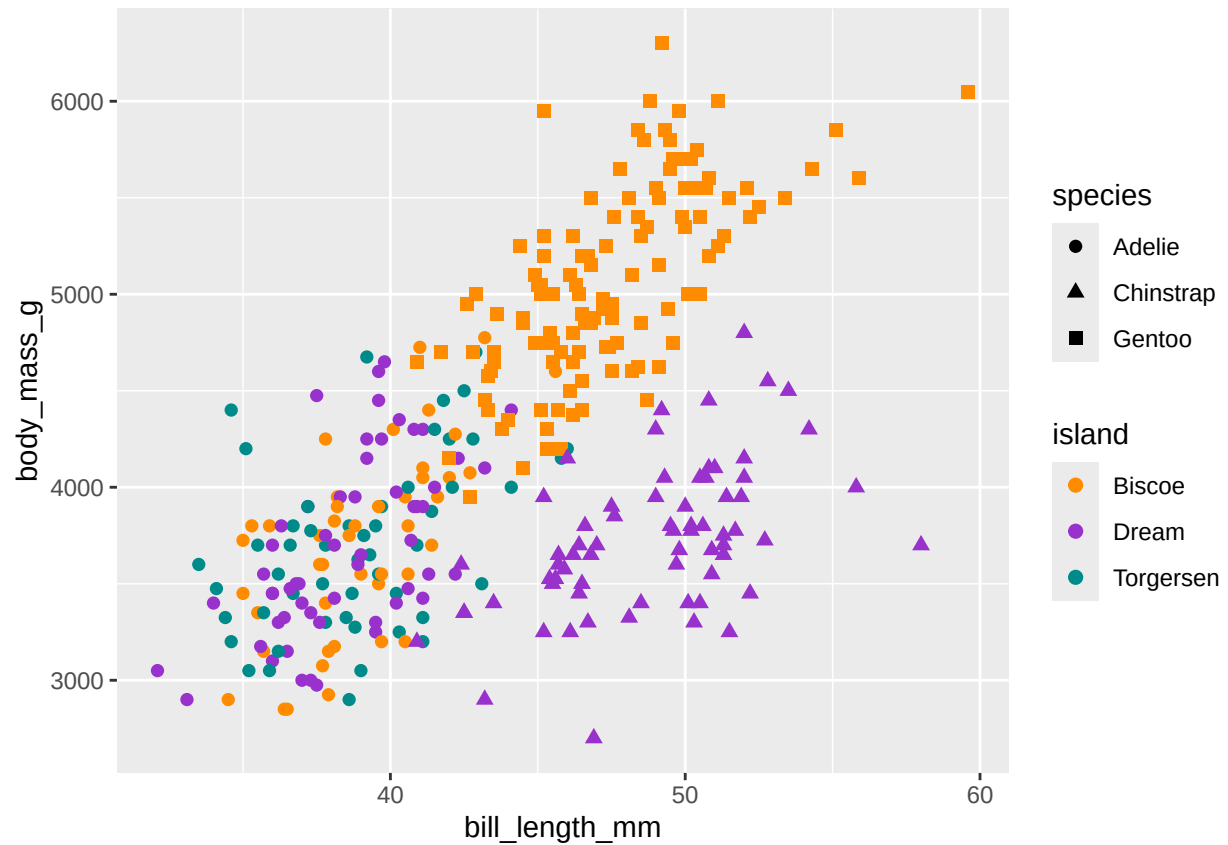
```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



Outro exemplo:

```
ggplot(
  data = penguins,
  aes(
    x = bill_length_mm,
    y = body_mass_g,
    color = island,
    shape = species
  )
) +
  geom_point(size = 2) +
  scale_color_manual(values = c("darkorange", "darkorchid", "cyan4"))
```

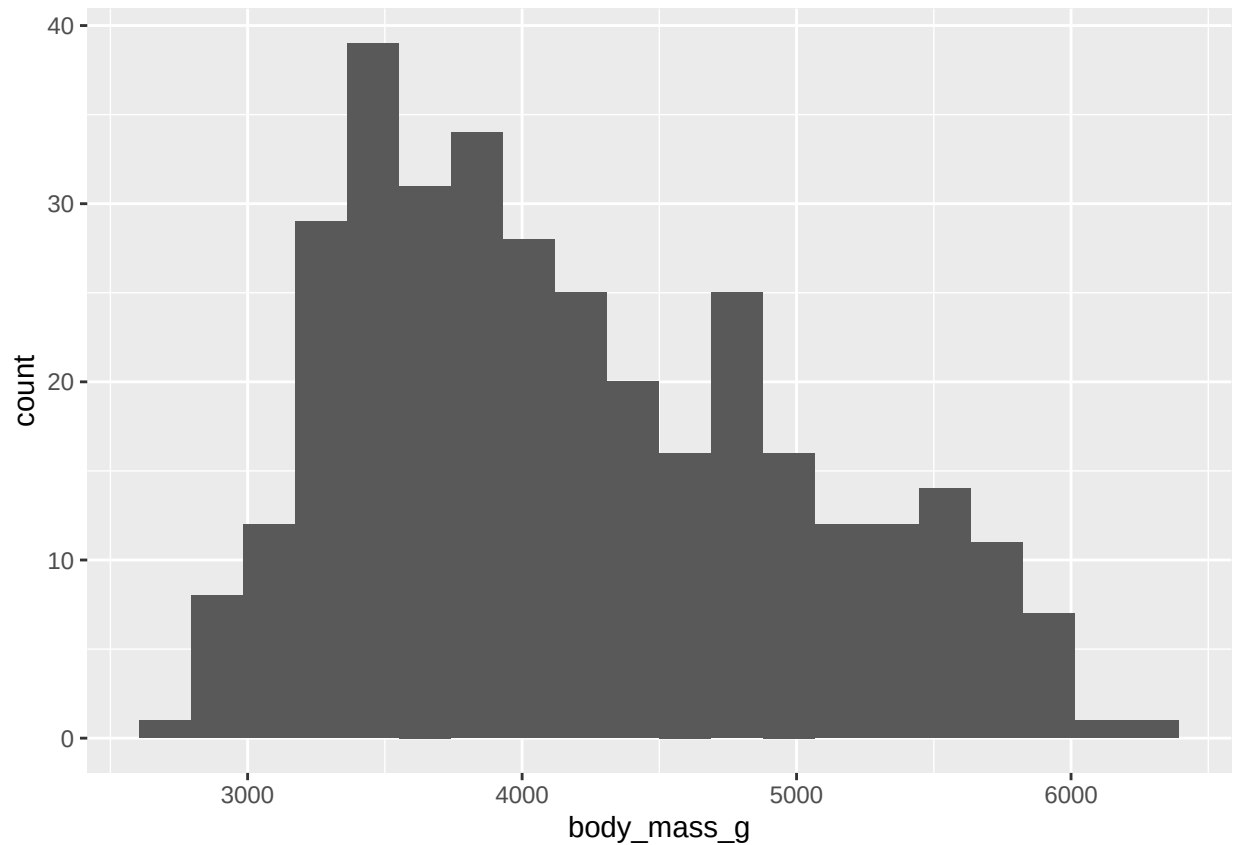
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Histogramas

```
ggplot(
  penguins,
  aes(
    x = body_mass_g
  )
) +
  geom_histogram(bins = 20)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_bin()').
```

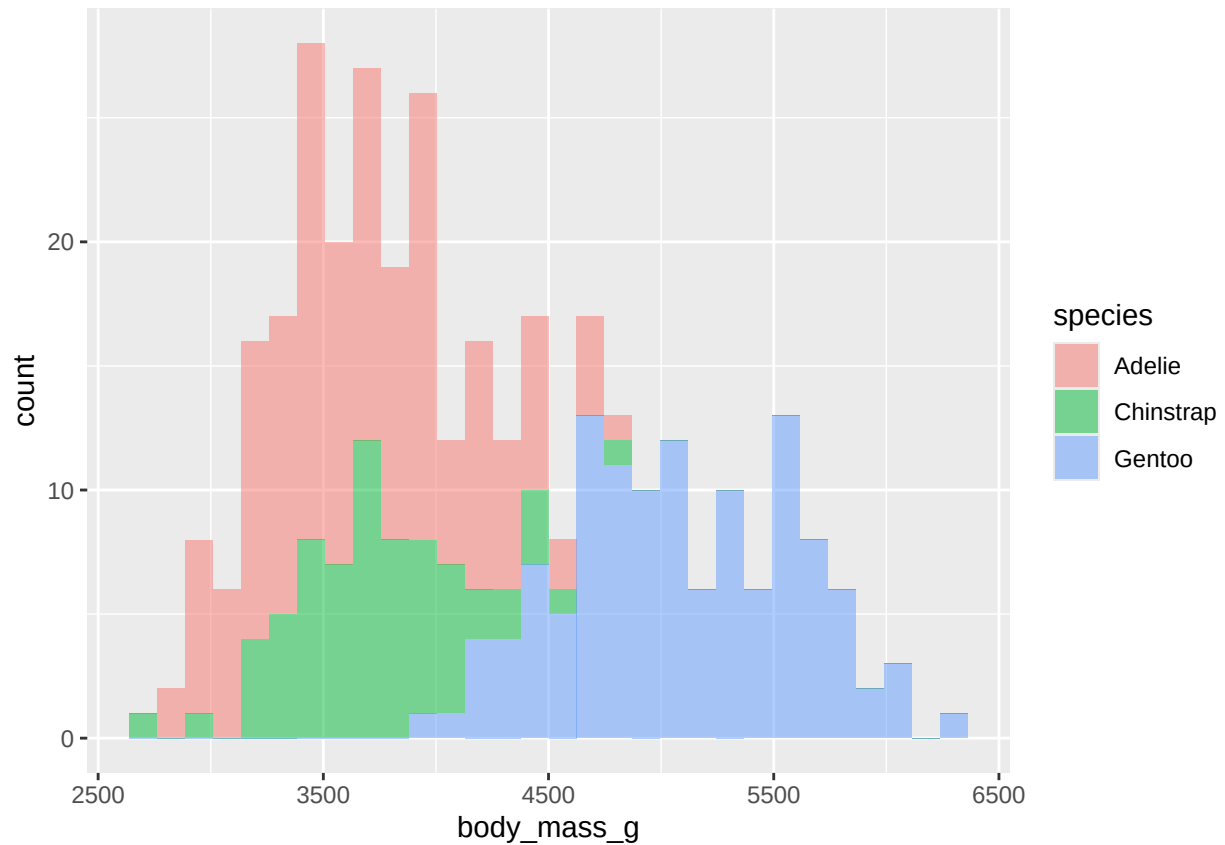



Agora diferenciando por espécie:

```
ggplot(  
  penguins,  
  aes(  
    x = body_mass_g,  
    fill = species  
  )  
) +  
  geom_histogram(alpha = 0.5)
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range  
## ('stat_bin()').
```



Outras maneiras de passar as informações também funcionam:

```
ggplot(
  penguins,
  aes(
    x = body_mass_g
  )
) +
  geom_histogram(alpha = 0.5, aes(fill = species))
```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

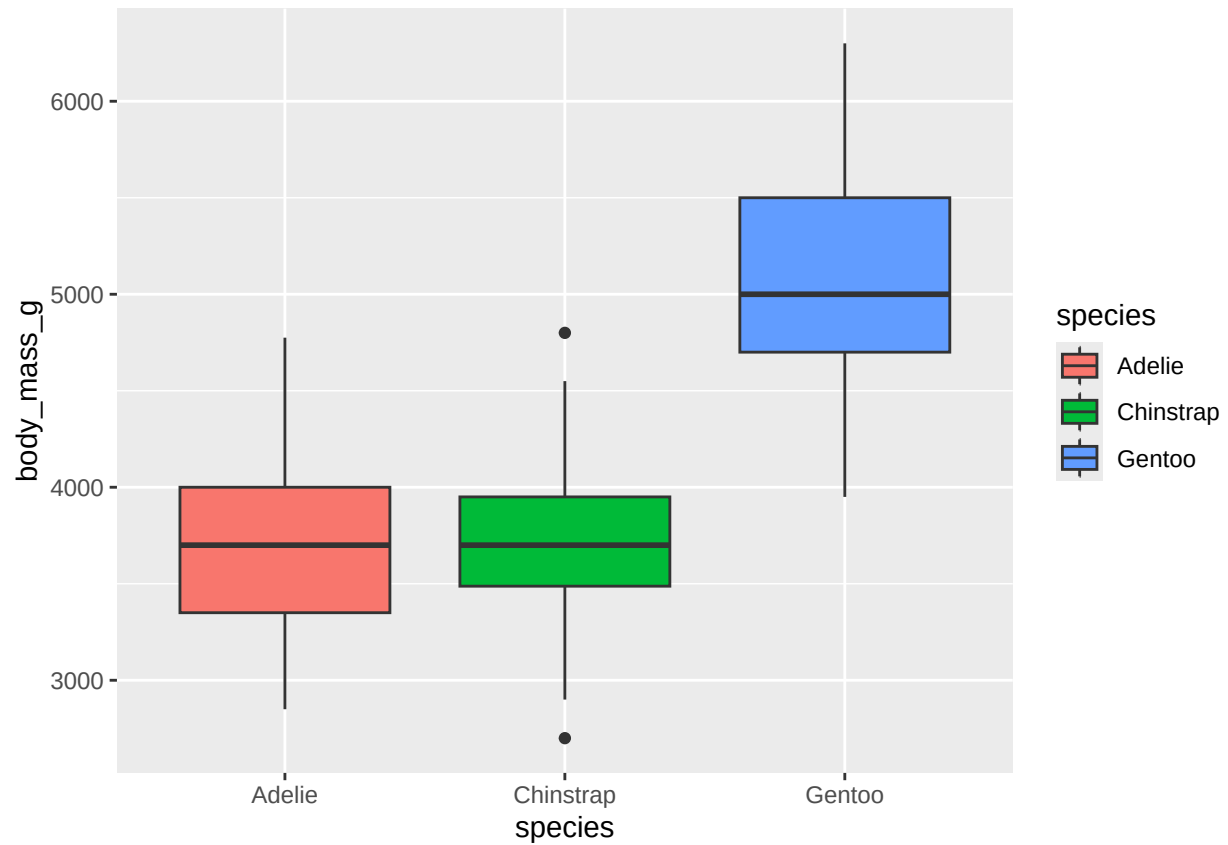
Warning: Removed 2 rows containing non-finite outside the scale range
('stat_bin()').



Boxplots

```
ggplot(
  penguins,
  aes(
    x = species,
    y = body_mass_g,
    fill = species
  )
) +
  geom_boxplot()
```

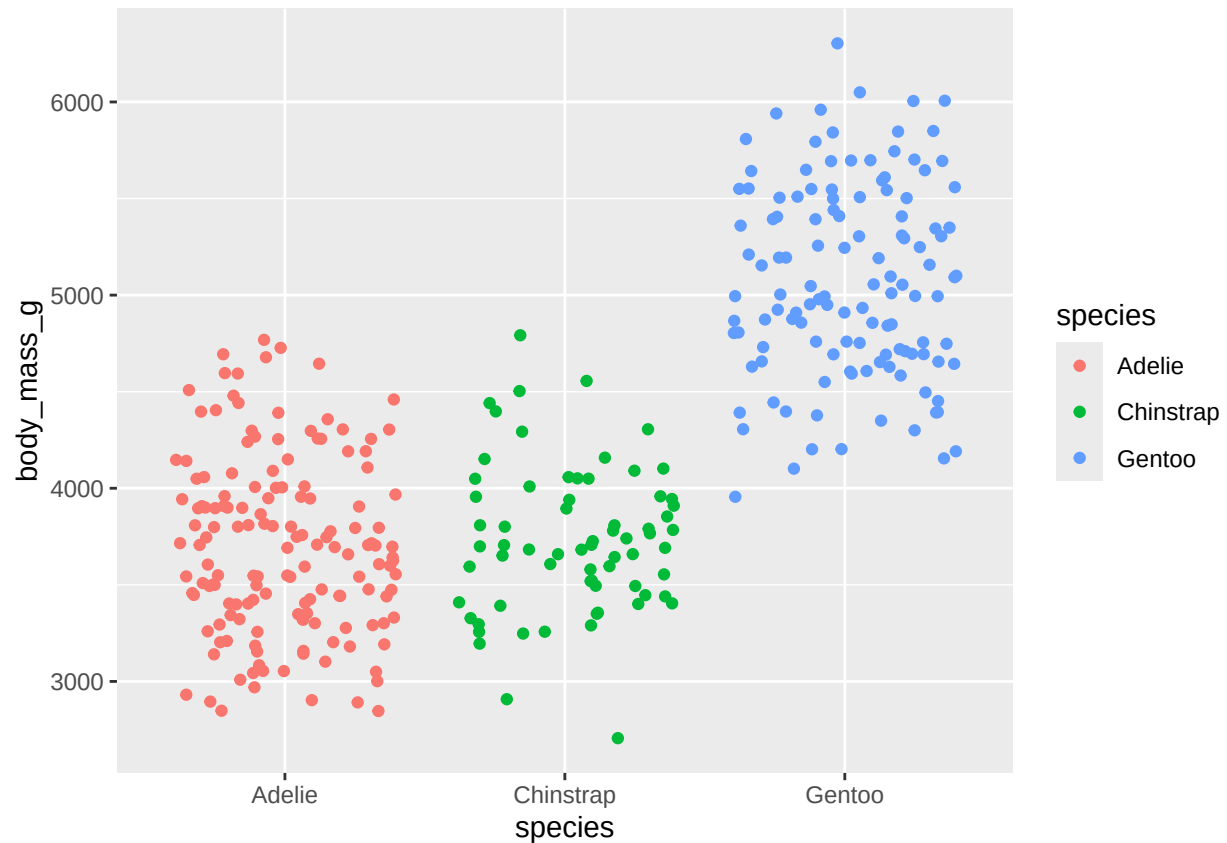
```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



Vendo a distribuição dos dados, em vez do *boxplot*:

```
ggplot(  
  penguins,  
  aes(  
    x = species,  
    y = body_mass_g,  
    color = species  
  )  
) +  
  geom_jitter()
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

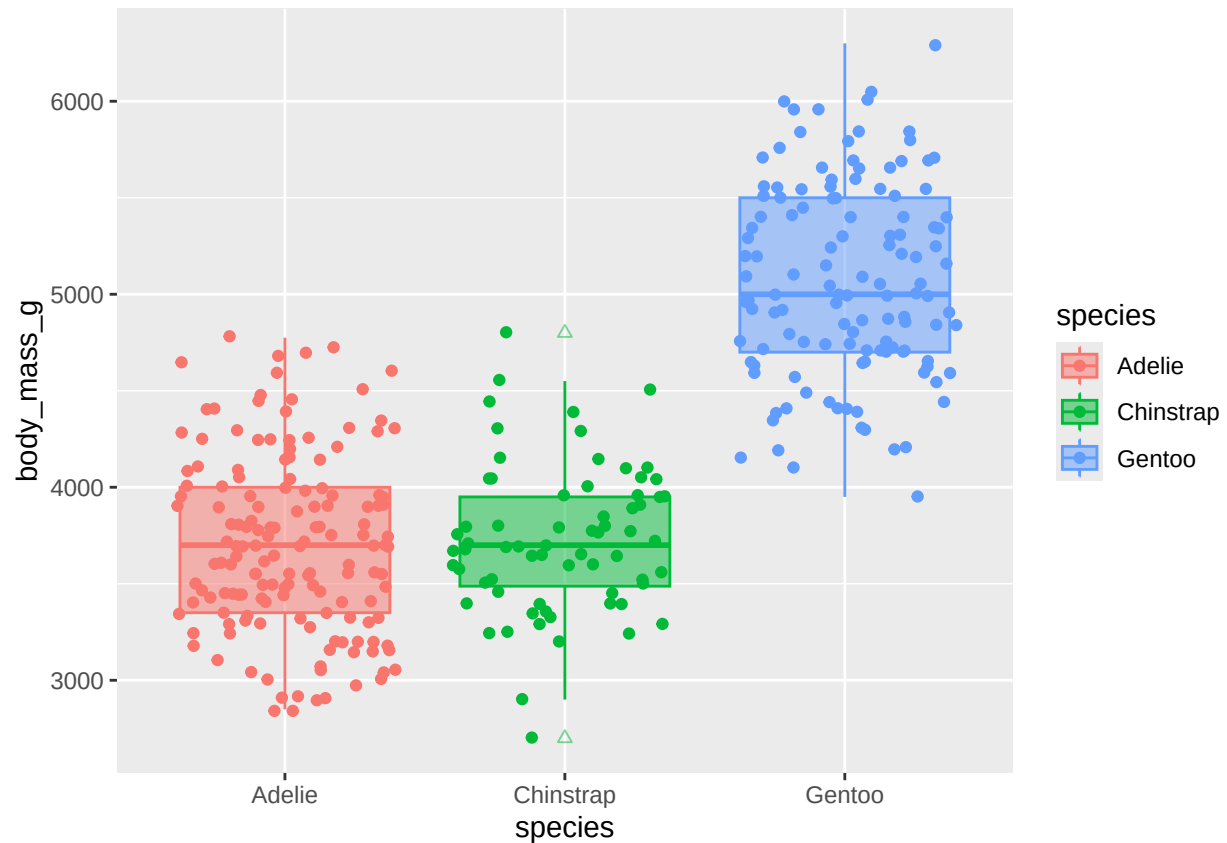


Juntando *boxplot* e *jitter*:

```
ggplot(
  penguins,
  aes(
    x = species,
    y = body_mass_g,
    fill = species,
    color = species
  )
) +
  geom_boxplot(alpha = 0.5, outlier.shape = 2) +
  geom_jitter()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

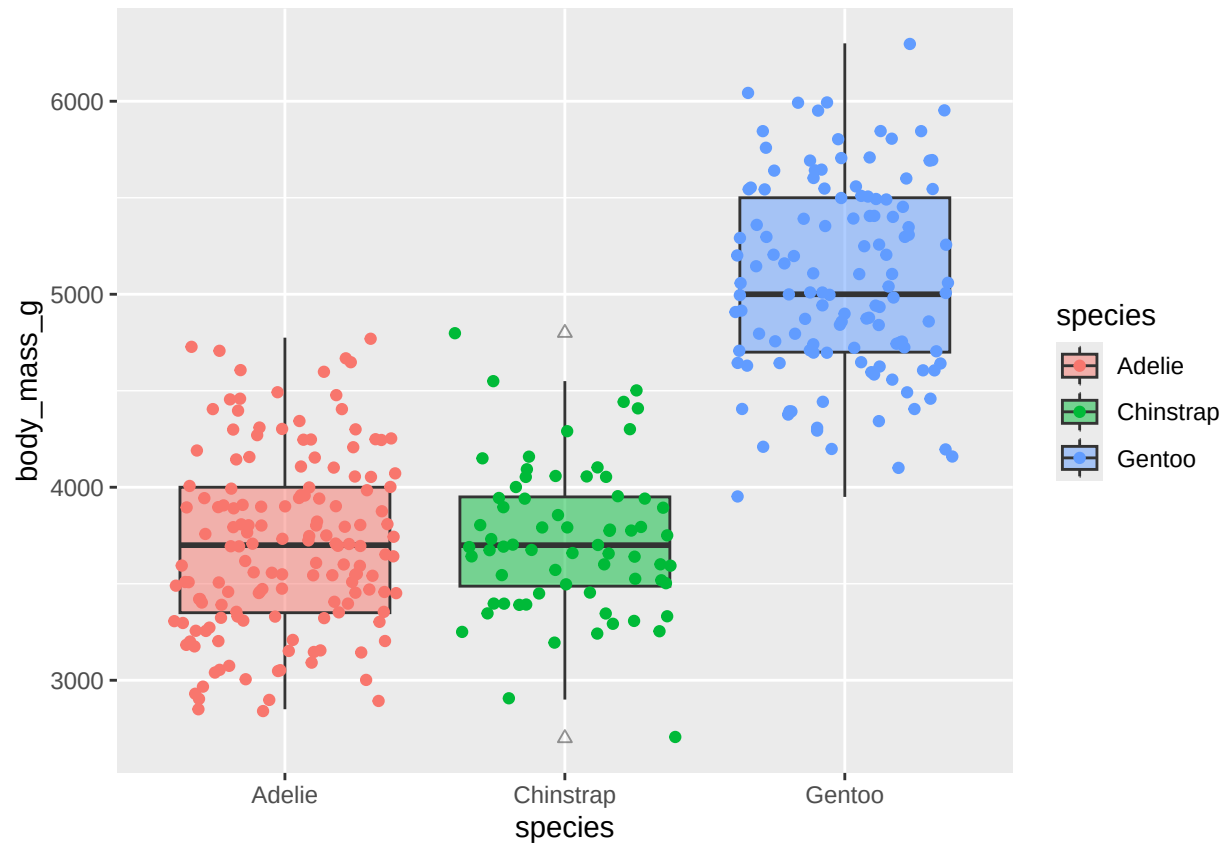


Os argumentos `fill` e `color` também podem ser passados apenas para os gráficos correspondentes. Faz até mais sentido, visualmente:

```
ggplot(
  penguins,
  aes(
    x = species,
    y = body_mass_g
  )
) +
  geom_boxplot(alpha = 0.5, outlier.shape = 2, aes(fill = species)) +
  geom_jitter(aes(color = species))
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



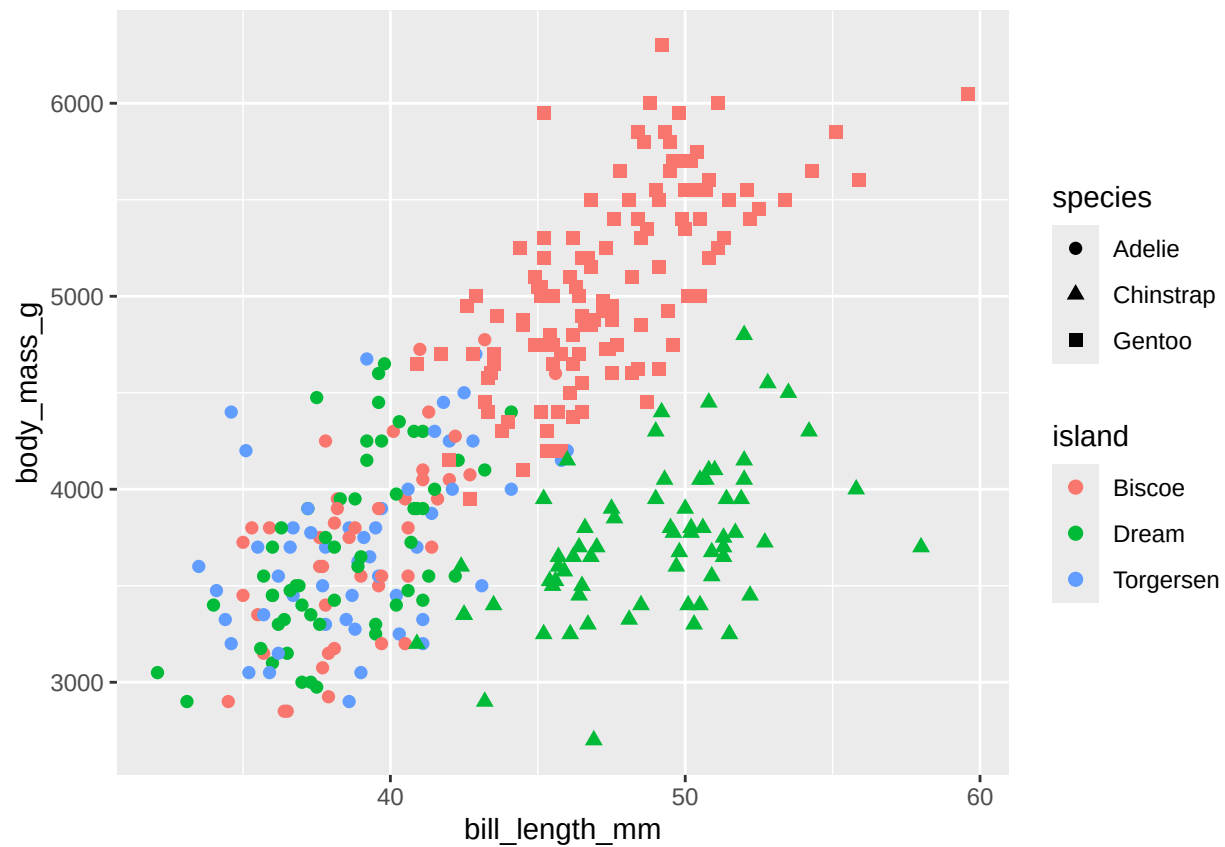
Facetas

Uma forma de dividir um gráfico em diversos gráficos de um jeito esperto.

Repare que aqui está confuso, temos cores e formatos diferentes para diferenciar ilha e espécie:

```
ggplot(
  data = penguins,
  aes(
    x = bill_length_mm,
    y = body_mass_g,
    color = island,
    shape = species
  )
) +
  geom_point(size = 2)
```

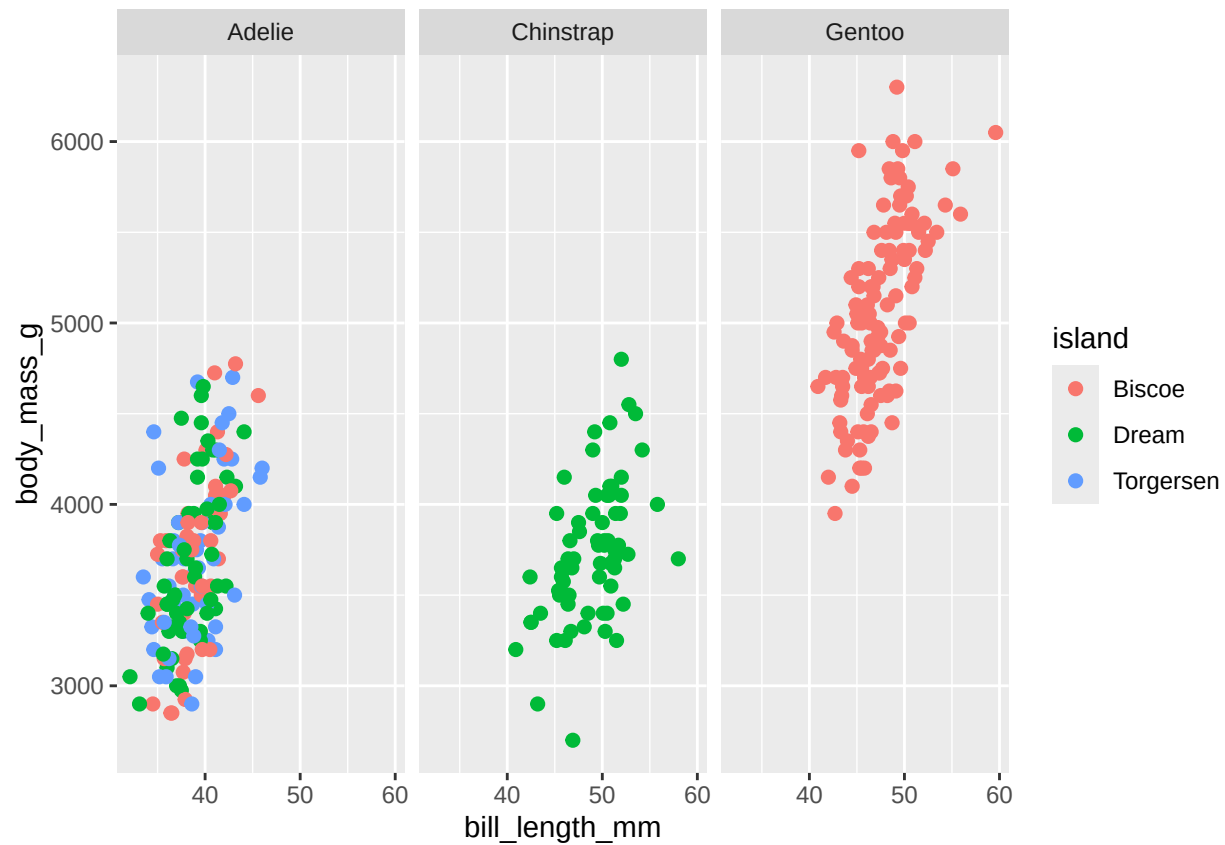
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Isso pode ser separado em diversos painéis, digamos, por espécie:

```
ggplot(
  data = penguins,
  aes(
    x = bill_length_mm,
    y = body_mass_g,
    color = island
  )
) +
  geom_point(size = 2) +
  facet_wrap(~species)
```

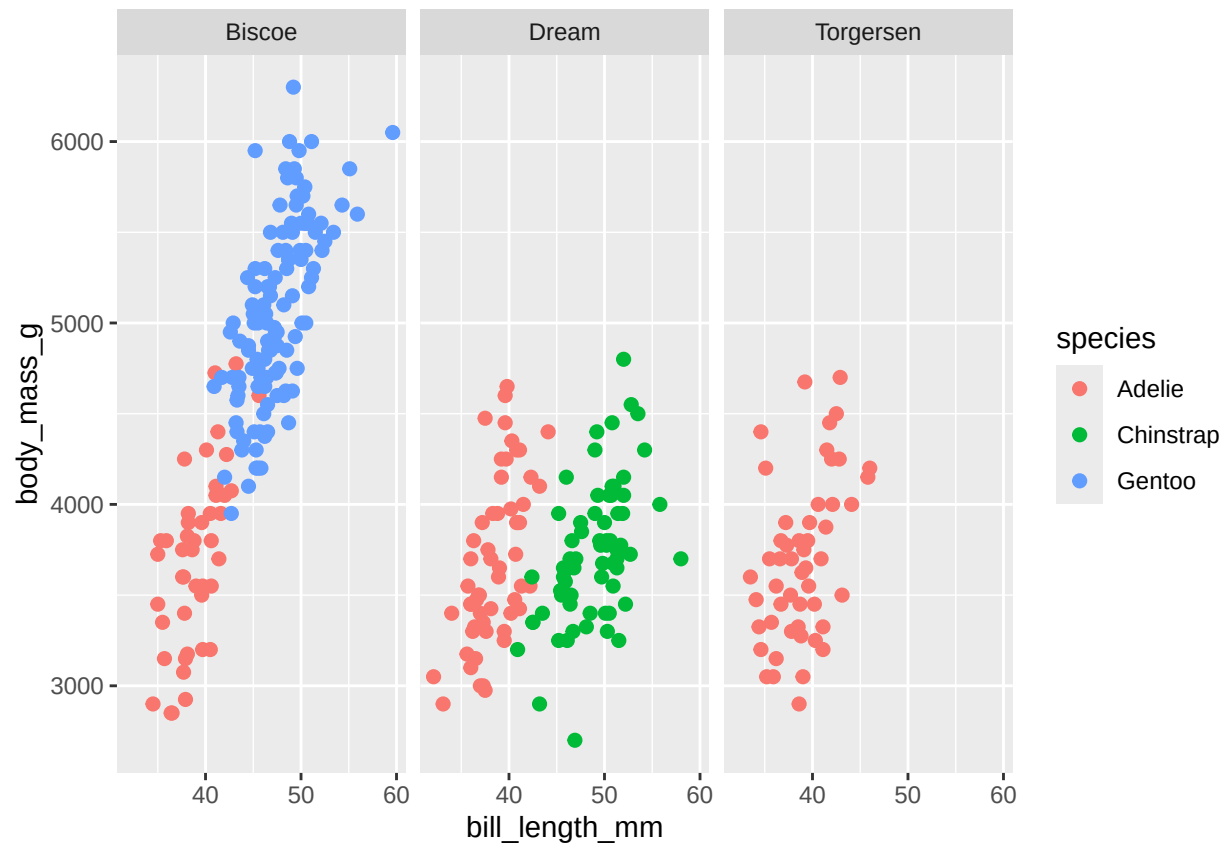
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Agora trocando: colorindo pela espécie e agrupando por ilha:

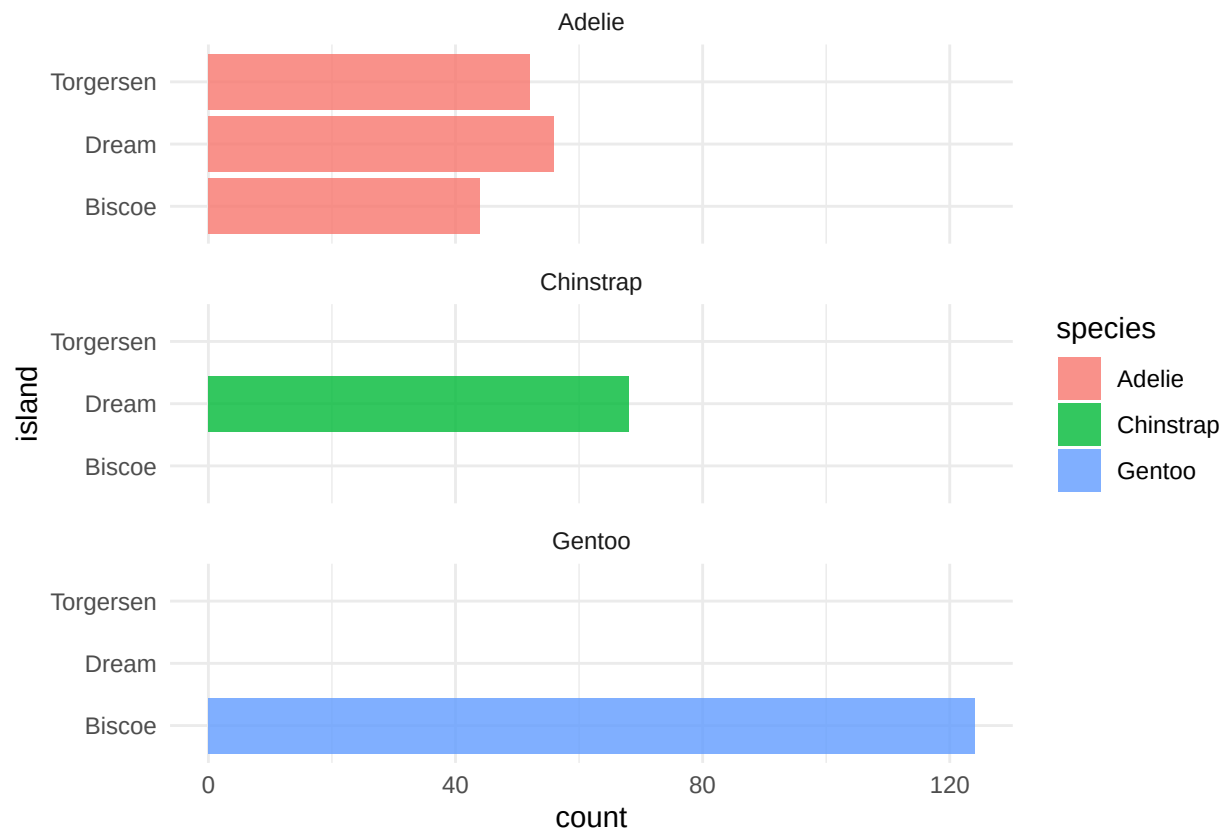
```
ggplot(
  data = penguins,
  aes(
    x = bill_length_mm,
    y = body_mass_g,
    color = species
  )
) +
  geom_point(size = 2) +
  facet_wrap(~island)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Mais um exemplo:

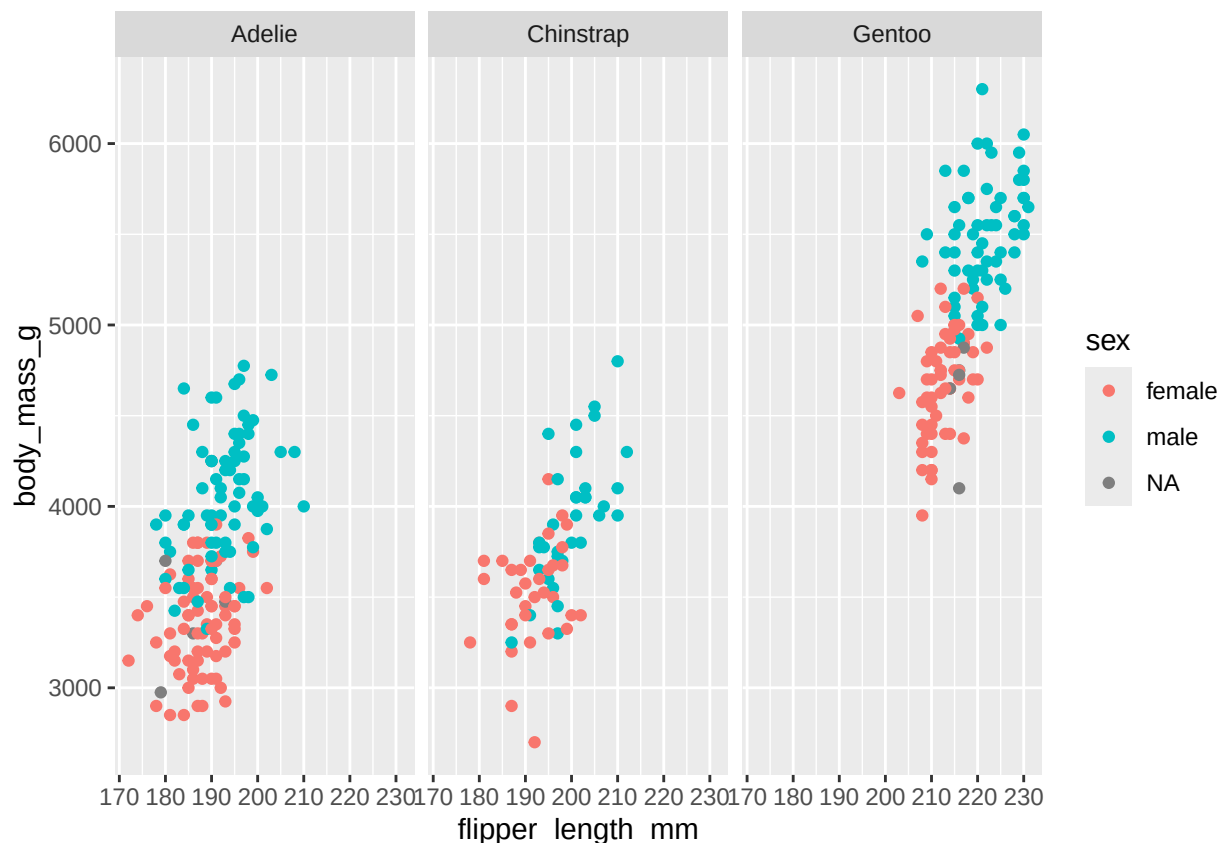
```
ggplot(
  penguins,
  aes(
    x = island,
    fill = species
  )
) +
  geom_bar(alpha = 0.8) +
  facet_wrap(~species, ncol = 1) +
  coord_flip() +
  theme_minimal()
```



Outro exemplo:

```
ggplot(
  penguins,
  aes(
    x = flipper_length_mm,
    y = body_mass_g,
    color = sex
  )
) +
  geom_point() +
  facet_wrap(~species)
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Juntando filtragem e visualização

```
penguins %>%
  select(species, body_mass_g, ends_with("_mm")) %>%
  ggpairs(aes(color = species))
```

```
## plot: [1, 1] [=>-----] 4% est: 0s
```

```
## plot: [1, 2] [==>-----] 8% est: 1s
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## ('stat_boxplot()').
```

```
## plot: [1, 3] [====>-----] 12% est: 1s
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## ('stat_boxplot()').
```

```
## plot: [1, 4] [=====>-----] 16% est: 2s
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## ('stat_boxplot()').
```

```

## plot: [1, 5] [=====>-----] 20% est: 2s

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_boxplot()').

## plot: [2, 1] [=====>-----] 24% est: 2s
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_bin()').

## plot: [2, 2] [=====>-----] 28% est: 2s

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_density()').

## plot: [2, 3] [=====>-----] 32% est: 2s

## Warning: Removed 2 rows containing missing values

## plot: [2, 4] [=====>-----] 36% est: 2s

## Warning: Removed 2 rows containing missing values

## plot: [2, 5] [=====>-----] 40% est: 2s

## Warning: Removed 2 rows containing missing values

## plot: [3, 1] [=====>-----] 44% est: 1s
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_bin()').

## plot: [3, 2] [=====>-----] 48% est: 1s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## plot: [3, 3] [=====>-----] 52% est: 1s

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_density()').

## plot: [3, 4] [=====>-----] 56% est: 1s

## Warning: Removed 2 rows containing missing values

## plot: [3, 5] [=====>-----] 60% est: 1s

```

```

## Warning: Removed 2 rows containing missing values

## plot: [4, 1] [=====>-----] 64% est: 1s
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_bin()').

## plot: [4, 2] [=====>-----] 68% est: 1s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## plot: [4, 3] [=====>-----] 72% est: 1s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## plot: [4, 4] [=====>-----] 76% est: 1s

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_density()').

## plot: [4, 5] [=====>-----] 80% est: 1s

## Warning: Removed 2 rows containing missing values

## plot: [5, 1] [=====>-----] 84% est: 0s
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_bin()').

## plot: [5, 2] [=====>-----] 88% est: 0s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## plot: [5, 3] [=====>-----] 92% est: 0s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

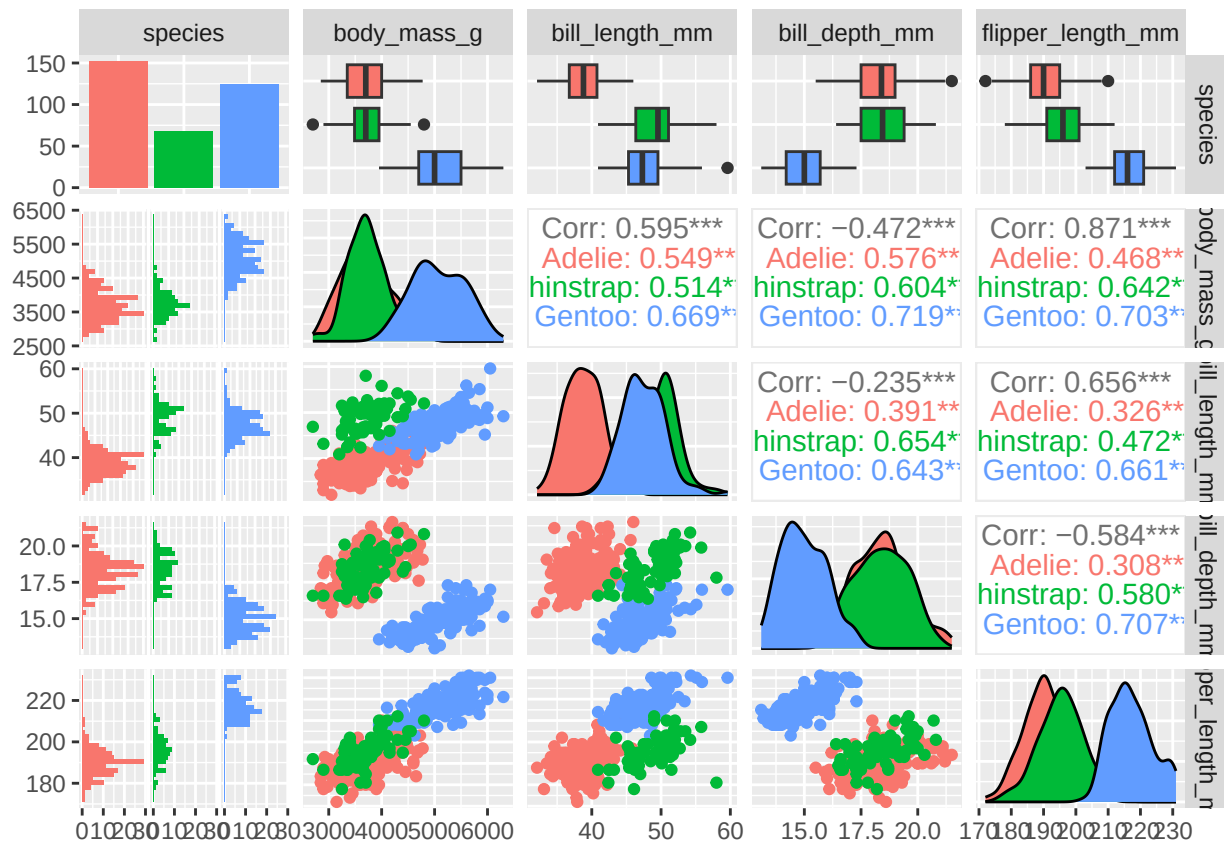
## plot: [5, 4] [=====>--] 96% est: 0s

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## plot: [5, 5] [=====] 100% est: 0s

```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_density()').
```



Dois exemplos finais

Pergunta: *Desconsiderando indivíduos cujo sexo não é conhecido, qual é a média dos pesos (em kg) dentro de cada espécie e sexo?*

```
penguins %>%
  filter(!is.na(sex)) %>%
  mutate(body_mass_kg = body_mass_g / 1000) %>%
  group_by(species, sex) %>%
  summarize(
    Média = mean(body_mass_kg, na.rm = TRUE)
  )
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by species and sex.
## i Output is grouped by species.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(species, sex))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
## # A tibble: 6 x 3
## # Groups:   species [3]
##   species sex Média
##   <fct>   <fct> <dbl>
## 1 Adelie female 3.37
## 2 Adelie male 4.04
## 3 Chinstrap female 3.53
## 4 Chinstrap male 3.94
## 5 Gentoo female 4.68
## 6 Gentoo male 5.48
```

Pergunta: *Desconsiderando indivíduos cujo sexo não é conhecido, como é a distribuição da razão entre as medidas do bico em relação ao peso (em g) dentro de cada espécie e sexo?*

```
penguins %>%
  filter(!is.na(sex)) %>%
  mutate(bill_ratio = bill_length_mm / bill_depth_mm) %>%
  ggplot(
    aes(
      x = body_mass_g,
      y = bill_ratio,
      color = species
    )
  ) +
  geom_point() +
  facet_wrap(~sex) +
  theme_minimal()
```